ENTITY RELATIONSHIP DIAGRAM



ERD (ENTITY RELATIONSHIP DIAGRAM)

- is modeling tool used to depict graphically a database design before it is actually implemented.
- It has three basic components, namely, an **Entity**, **Relationship** and an **Attribute**.

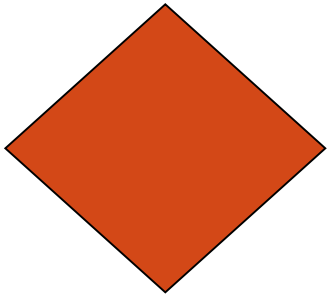
ENTITY

- is a real-world object distinguishable or unique from other objects.
- An entity can be a **concrete** or **physical** object like *employee*, *student*, *faculty*, *customer* etc. Or it could also be **conceptual** or **abstract** like *transaction*, *order*, *course*, *subjects* etc.
- It can be thought of as a noun like *student, employee* etc.
- It is normally represented by a rectangle shape.

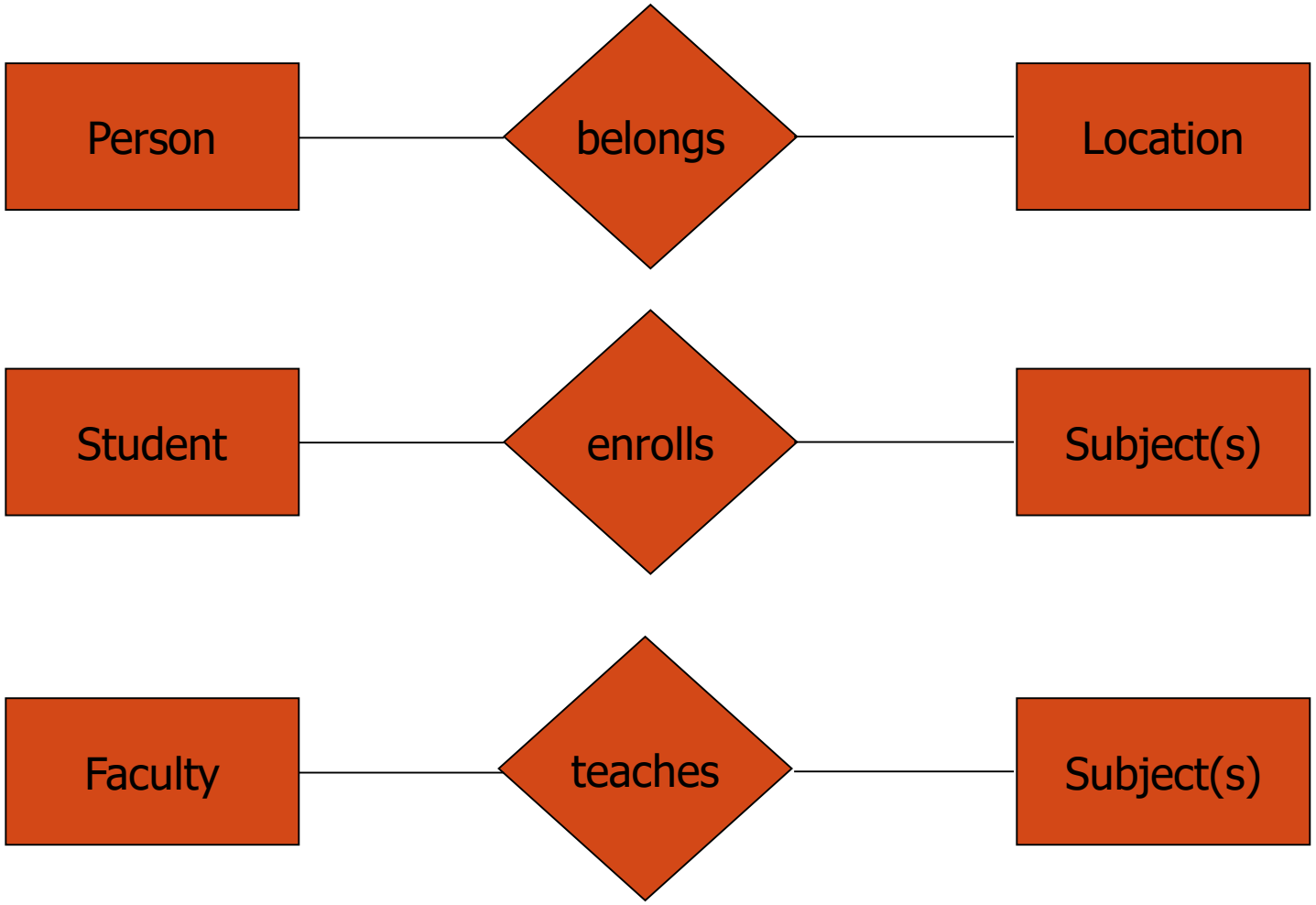


RELATIONSHIP

- is a way of relating one entity to another. Entities can therefore participate in a relationship.
- it is commonly thought as a *verb* connecting the *entities* or *nouns.*
- It is normally represented by a diamond shape.



EXAMPLE OF ENTITIES WITH RELATIONSHIP

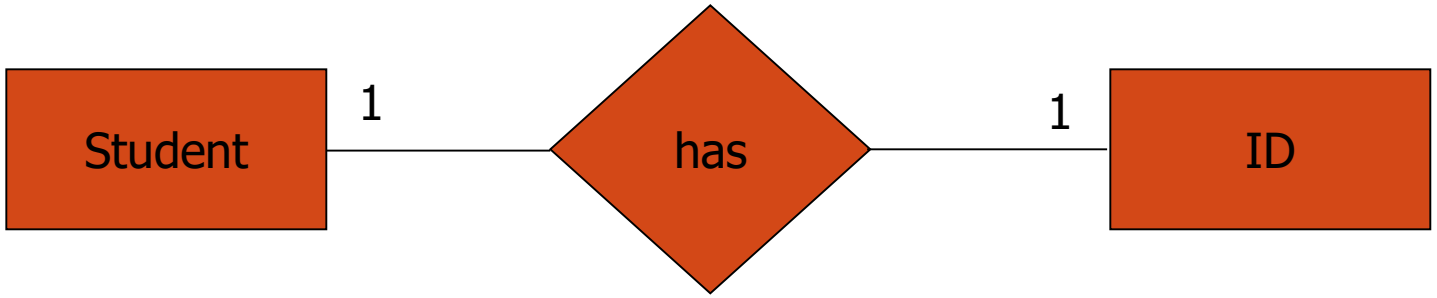


TYPES OF RELATIONSHIP

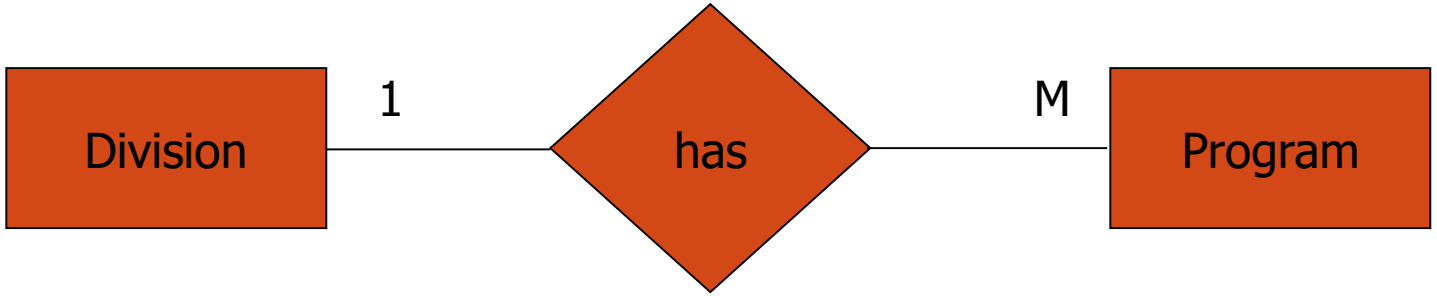
- One-to-One
- One-to-Many
- Many-to-Many

TYPES OF RELATIONSHIP

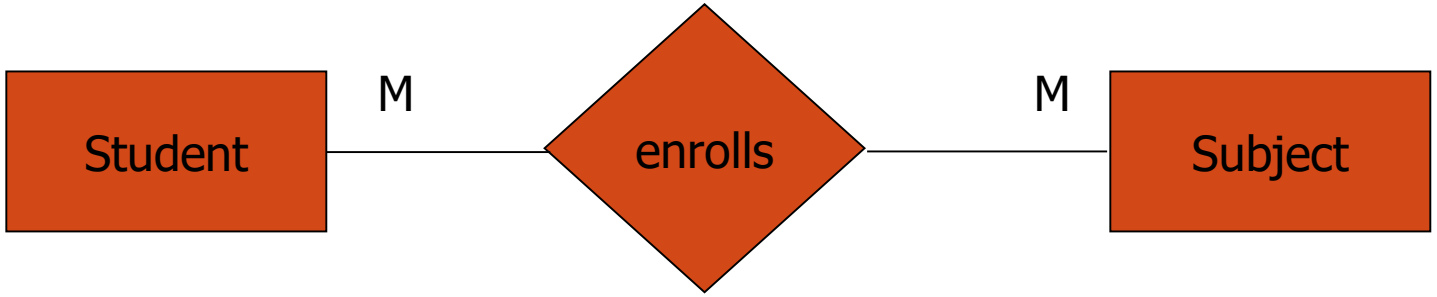
One-to-One



One-to-Many

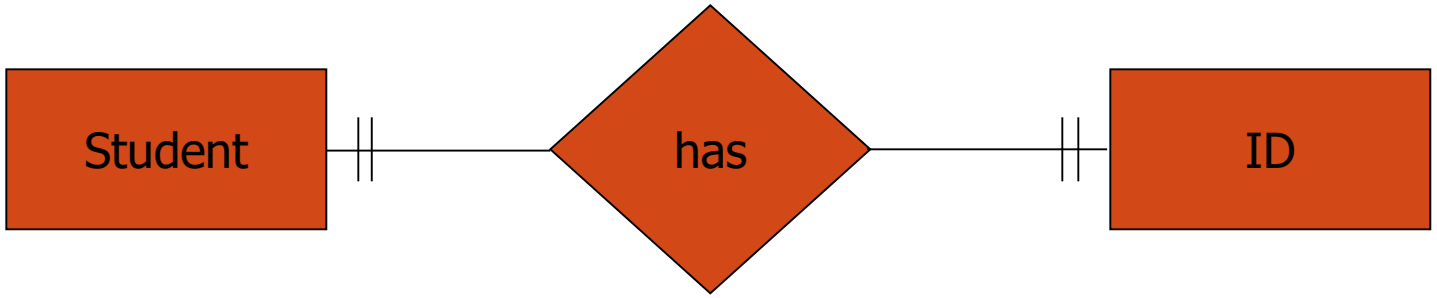


Many-to-Many

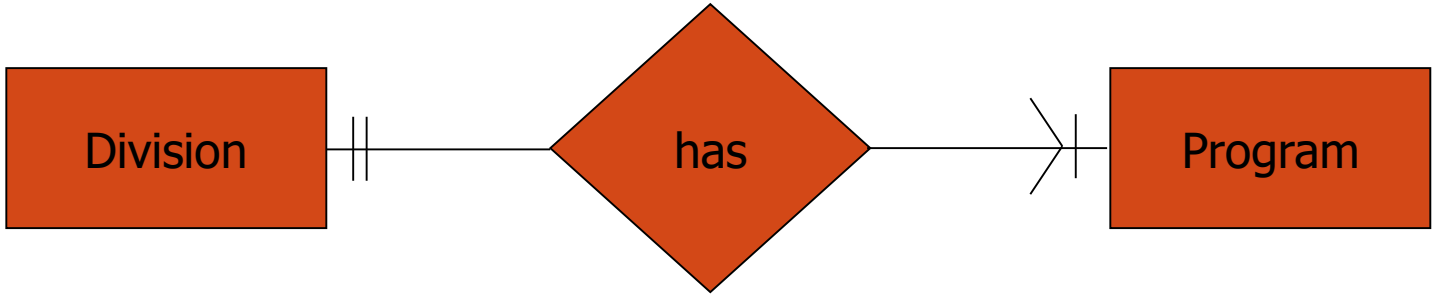


TYPES OF RELATIONSHIP (CROW NOTATION)

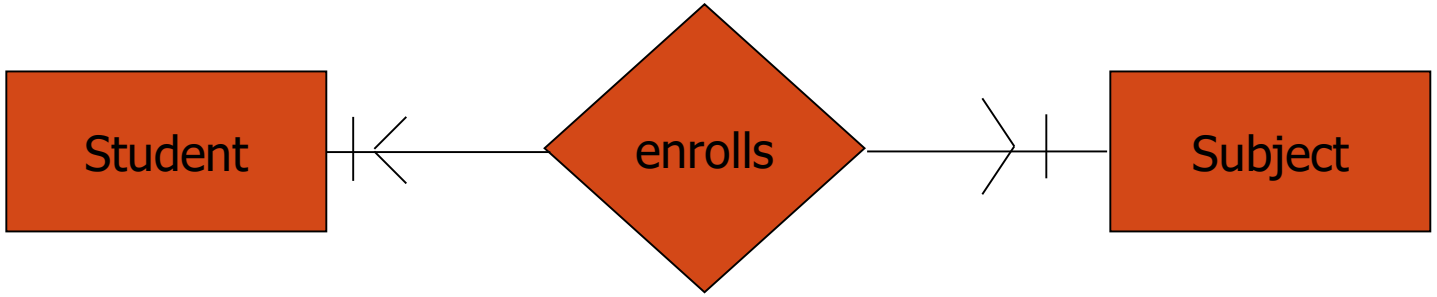
One-to-One



One-to-Many

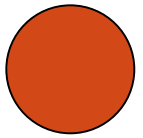


Many-to-Many

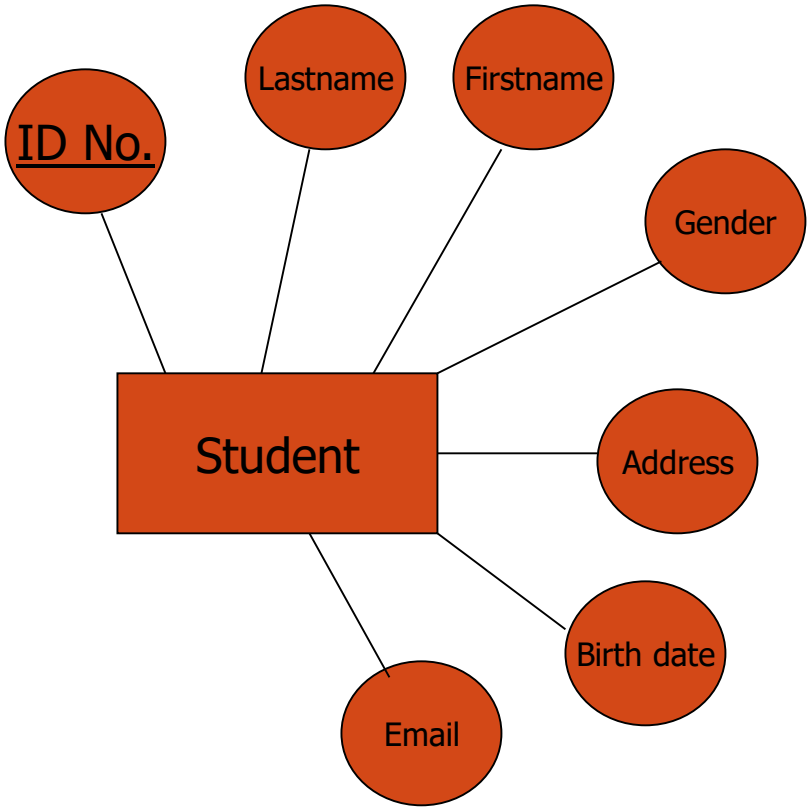


ATTRIBUTE

- Refers to the characteristic or basic fact or field of an Entity or Relationship.
- For example a Student entity could have the following **attributes** *ID Number*, *Last Name, First Name, Address, Birth Date* etc.
- A relationship could also have an attribute for example an Entity name **Student** enrolls (relationship) to a **Course/Program**. Now, when you enroll you enroll on a certain date so you will have an attribute of *Enrollment Date* under **Enroll** relationship.
- It is normally represented by a **circle**.

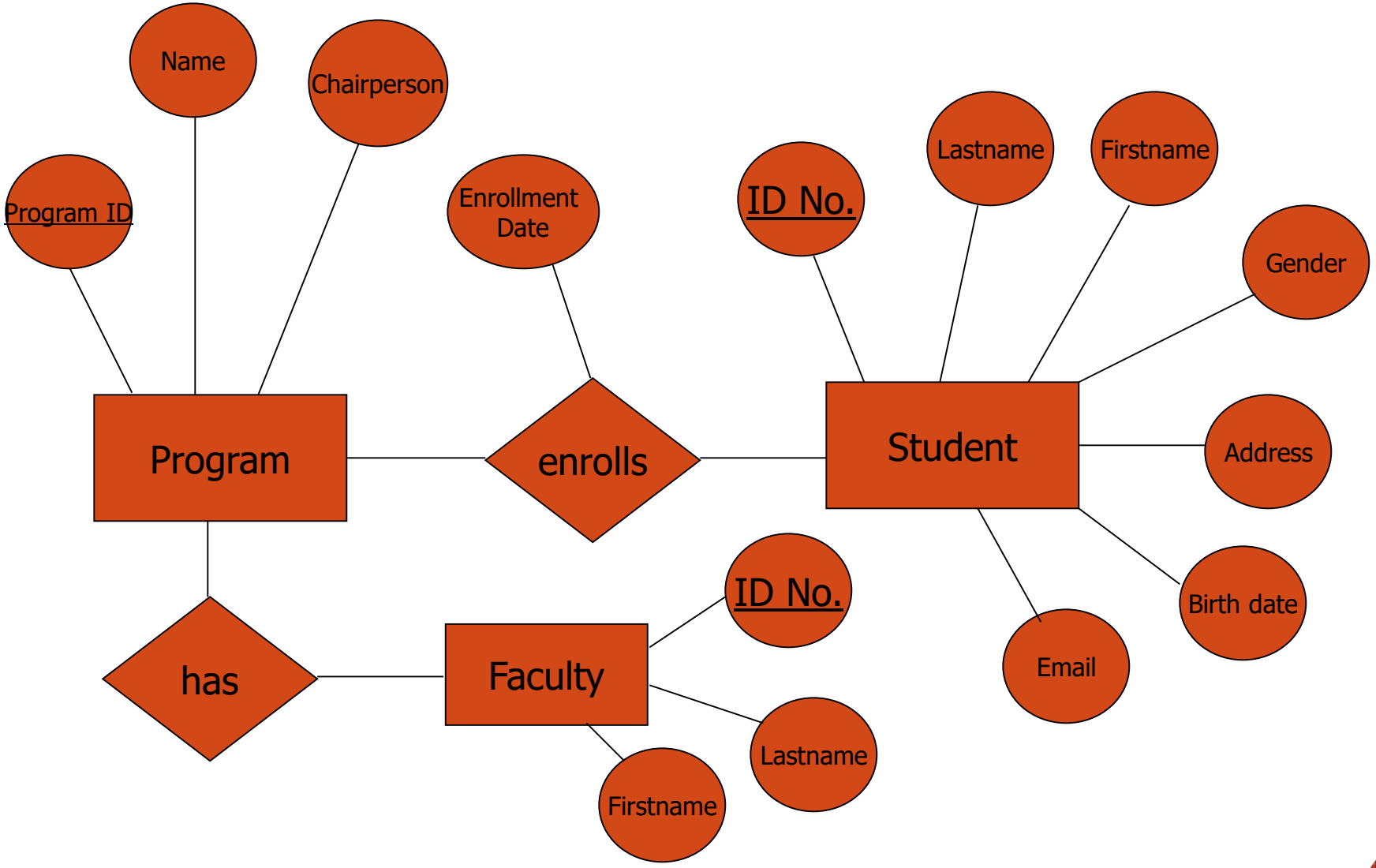


EXAMPLE OF ATTRIBUTES

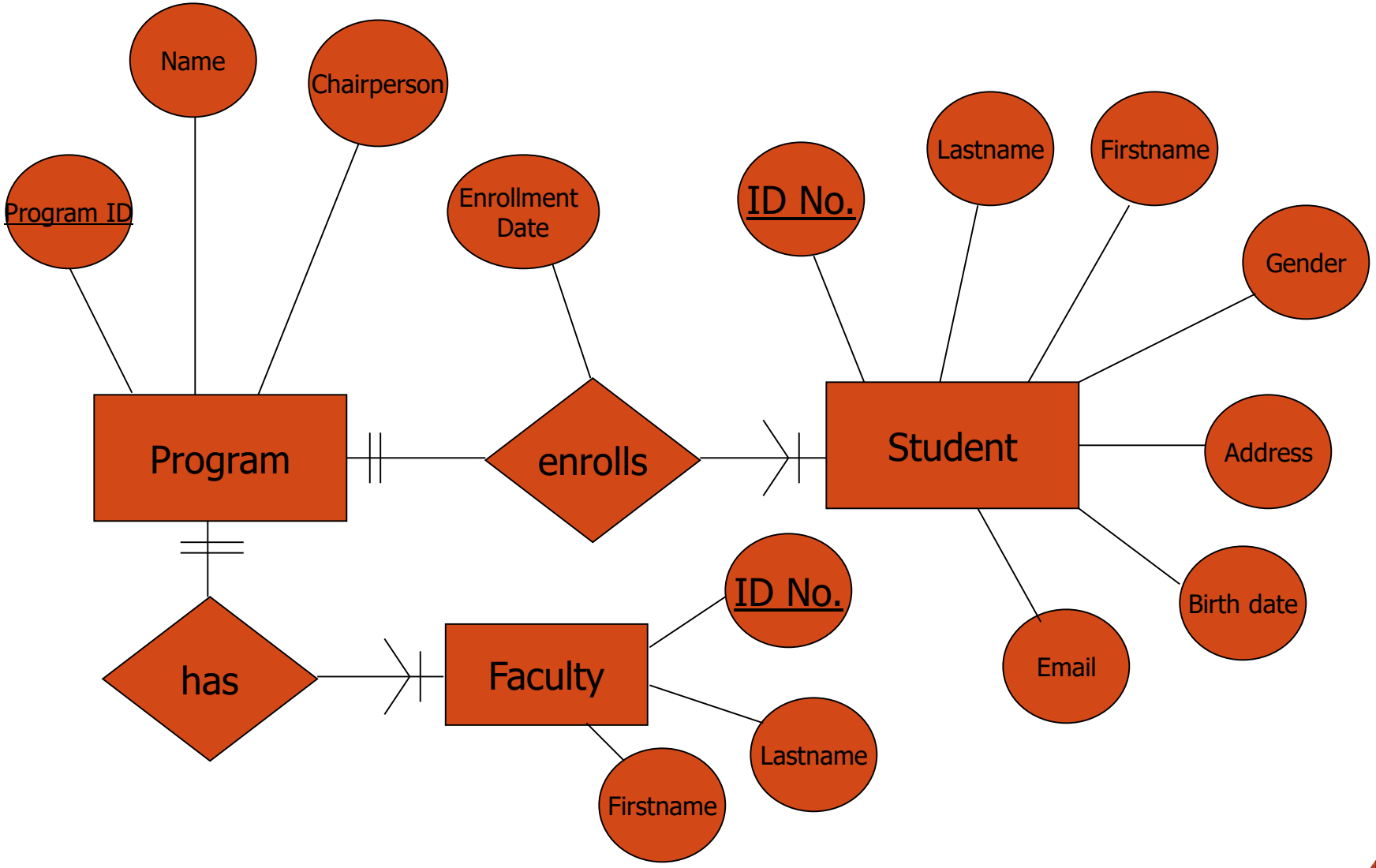


Note : A Primary Key is Underlined, in this case the ID No.

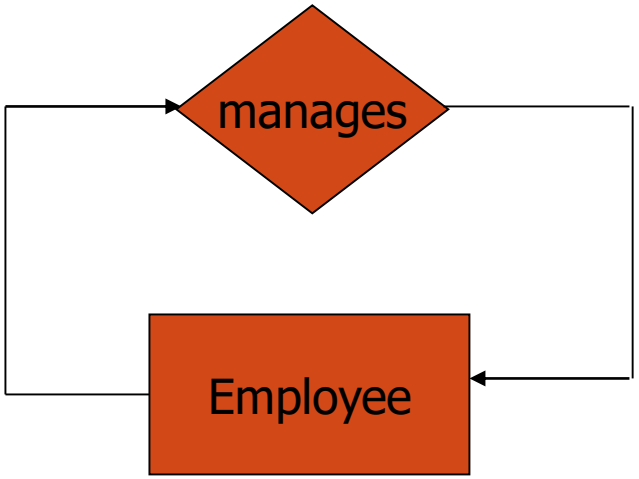
EXAMPLE OF ATTRIBUTES



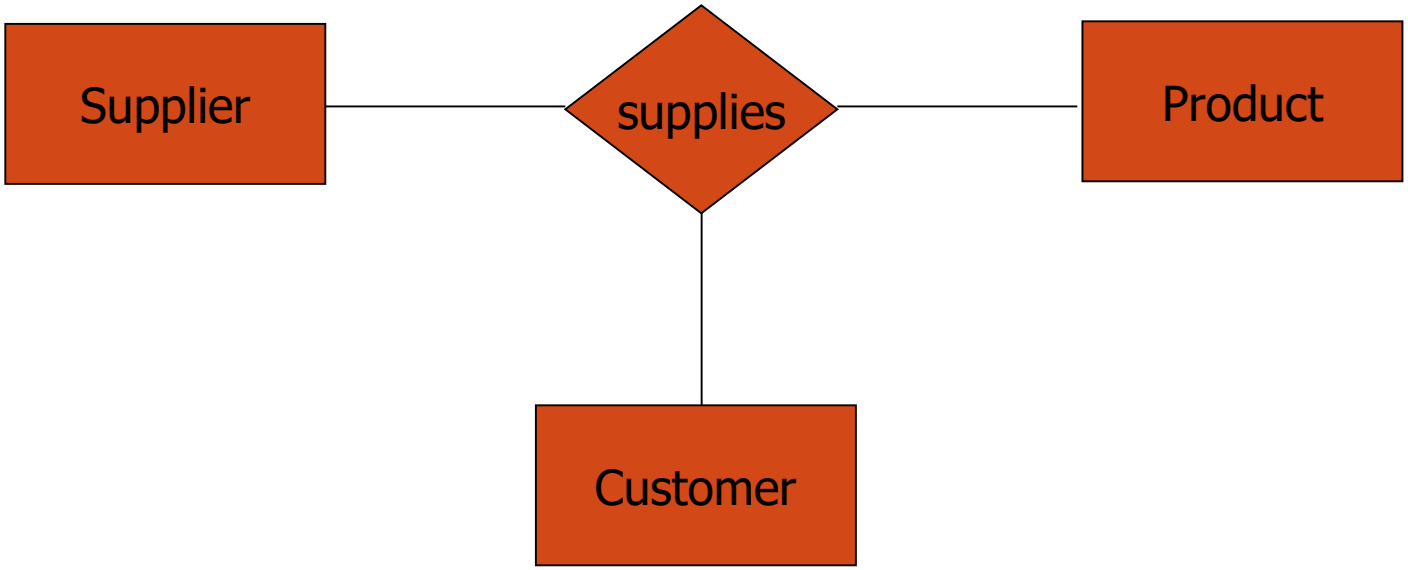
EXAMPLE OF ATTRIBUTES



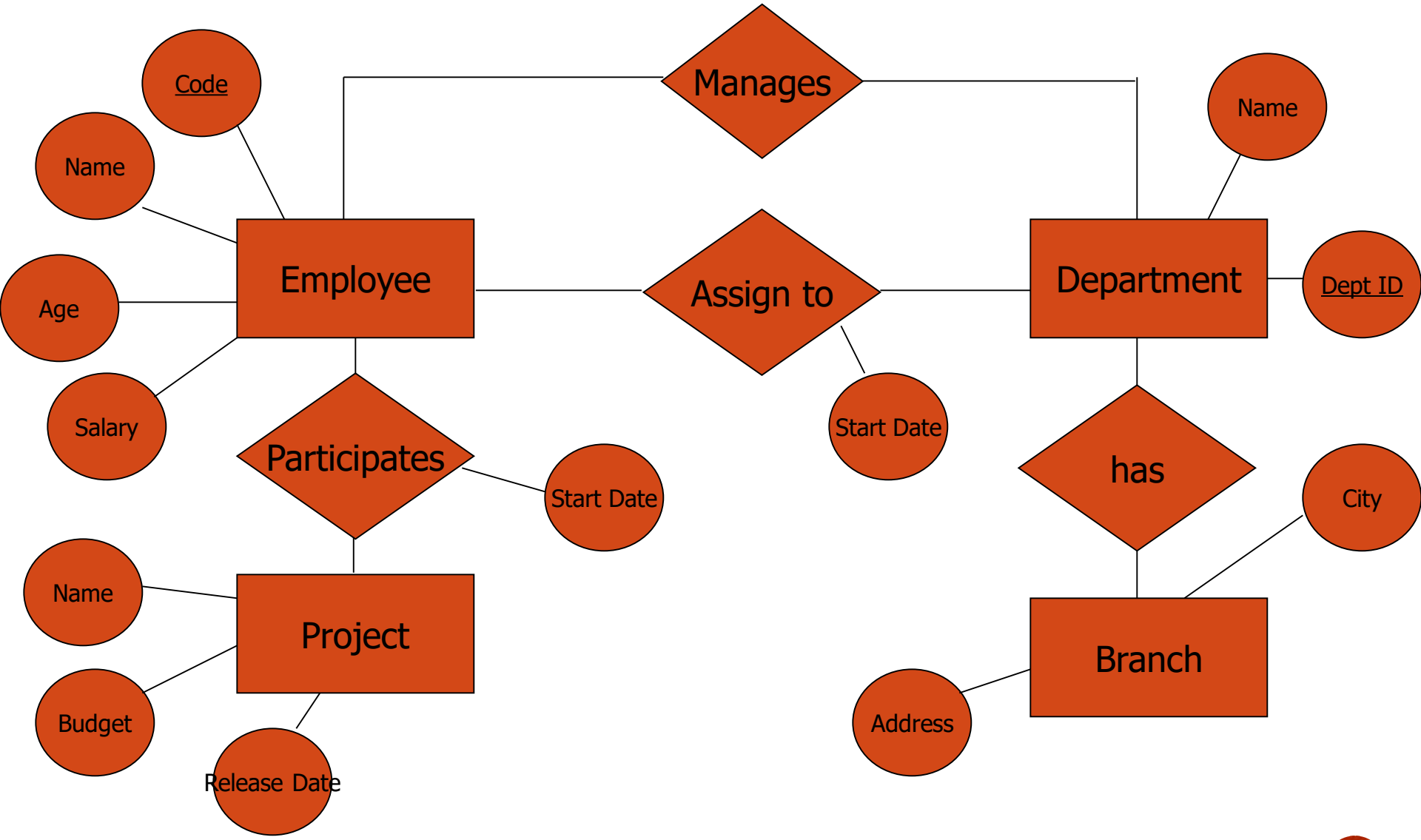
RECURSIVE ERD



ERD WITH TERNARY RELATIONSHIP



ERD (MORE EXAMPLES)



LOGICAL OPERATORS



LOGICAL OPERATORS

- The Logical operators are those that are true or false. They return a true or false values to combine one or more true or false values.



Operator	Description
AND	Logical AND compares between two Booleans as expression and returns true when both expressions are true…
OR	Logical OR compares between two Booleans as expression and returns true when one of the expression is true…
NOT	Not takes a single Boolean as an argument and changes its value from false to true or from true to false….



Special operators		
Operator	Description	Operates on
IN	The IN operator checks a value within a set of values separated by commas and retrieve the rows from the table which are matching....	Any set of values of the same datatype
BETWEEN	The SQL BETWEEN operator tests an expression against a range. The range consists of a beginning, followed by an AND keyword and an end expression....	Numeric, characters, or datetime values
ANY	ANY compares a value to each value in a list or results from a query and evaluates to true if the result of an inner query contains at least one row....	A value to a list or a single - columns set of values
ALL	ALL is used to select all records of a SELECT STATEMENT. It compares a value to every value in a list or results from a query. The ALL must be preceded by the comparison operators and evaluates to TRUE if the query returns no rows....	A value to a list or a single - columns set of values
SOME	SOME compare a value to each value in a list or results from a query and evaluate to true if the result of an inner query contains at least one row...	A value to a list or a single - columns set of values
EXISTS	The EXISTS checks the existence of a result of a subquery. The EXISTS subquery tests whether a subquery fetches at least one row. When no data is returned then this operator returns 'FALSE'...	Table



EXAMPLES

- SELECT cust_code, cust_name, cust_city,cust_country,grade FROM customer WHERE cust_country = 'UK' AND grade = 2;
- SELECT cust_code,cust_name, cust_city,cust_country,grade FROM customer WHERE cust_country = 'USA' OR grade = 3;
- SELECT * FROM customer WHERE NOT grade>1;
- SELECT cust_code,cust_name,cust_city,cust_country,grade FROM customer WHERE cust_country='UK' AND cust_city='London' AND grade>1;



RELATIONAL OPERATORS



RELATIONAL OPERATORS

- A comparison (or relational) operator is a mathematical symbol which is used to compare two values.
- Relational operators are used in conditions that compares one expression with another. The result of a comparison can be TRUE, FALSE, or UNKNOWN (an operator that has one or two NULL expressions returns UNKNOWN).



Operator	Description	Operates on
=	Equal to.	Any compatible data types
>	Greater than.	Any compatible data types
<	Less than.	Any compatible data types
>=	Greater than equal to.	Any compatible data types
<=	Less than equal to.	Any compatible data types
<>	Not equal to.	Any compatible data types



EXAMPLES

- SELECT * FROM agents WHERE commission < 0.12;
- SELECT * FROM agents WHERE commission >= 0.14;
- SELECT * FROM agents WHERE commission = 0.15;
- SELECT * FROM agents WHERE commission <> 0.15;



WILDCARD OPERATORS



WILDCARD OPERATORS

- The SQL wildcards can be used to search data within a table.
- SQL wildcards are used with SQL LIKE operator.
- The boolean NOT operator in the select statement can be used as wildcard NOT LIKE operator.



LIKE OPERATOR

- LIKE is the ANSI/ISO standard operator for comparing a column value to another column value, or to a quoted string. Returns either 1 (TRUE) or 0 (FALSE)
- The SQL LIKE operator is only applied on a field of types CHAR or VARCHAR to match a pattern.
- To match a pattern from a word, special characters, and wildcards characters may have used with LIKE operator.
- The LIKE operator can be used within any valid SQL statement, such as SELECT, INSERT INTO, UPDATE or DELETE.



Wildcards	Description
%	The percent sign character (%) represent a sequence of 0 (zero) or more characters.
Underscore (_)	The underscore character (_) represents a single character.
[charlist]	It represents any single character within a charlist
[^charlist] or [!charlist]	It represents any single character other than the charlist



EXAMPLES

- Refer to your hand outs

